

TCP/IP Core Protocols

- **TCP/IP** is a collection of communication protocols used to transfer data across networks.
- It includes protocols like **TCP, IP, UDP, and ARP**.
- **TCP/IP is open and routable**, meaning it can be used on different types of networks and devices.
- Each layer in the **TCP/IP model adds a header** to the data from the layer above (this is called **encapsulation**).

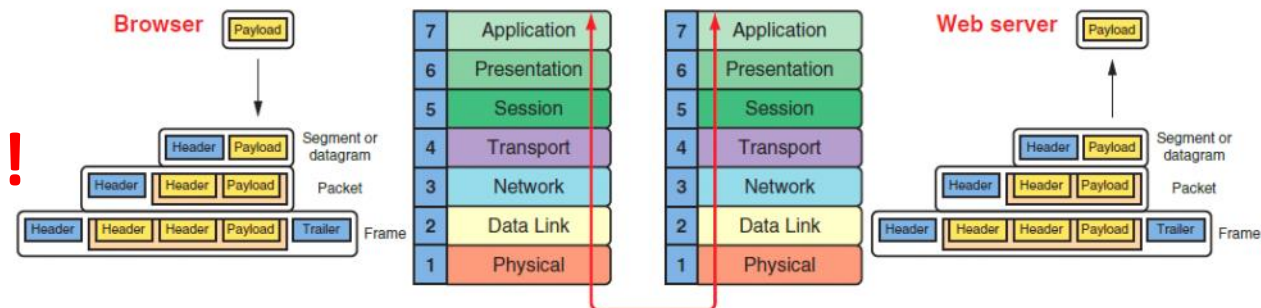


Figure 4-1 Each layer adds its own data and addresses its transmission to the corresponding layer in the destination device

Summary of TCP/IP Core Protocols

The TCP/IP model has different layers that help send data from one computer to another. Each layer adds some information to the data. Here's a simple explanation:

- **Layers 7, 6, and 5 (Application Layers):**

These layers create the **data**, also called the **payload**. For example, a browser makes the data when you visit a website.

- **Layer 4 – Transport Layer:**

This layer **usually uses TCP or UDP**. It adds a **header** with a **port number** to show where the data should go (which app).

The message is now called a **segment** (TCP) or **datagram** (UDP).

- **Layer 3 – Network Layer:**

This layer adds an **IP address** of the destination. Now the message is called a **packet**.

- **Layer 2 – Data Link Layer:**

This layer adds another **header and trailer**. It uses **physical addresses (MAC address)** to find the correct device in the local network. The message is now a **frame**.

- **Layer 1 – Physical Layer:**

This layer sends the frame over the **physical network** (like cables or Wi-Fi).

TCP (Transmission Control Protocol)

- TCP works in the **Transport layer** of the OSI model.
- It makes sure there is a stable **connection** before sending data (called a **three-way handshake**).
- It uses **sequence numbers and checksums** to check if the data is correct and in order.
- **Flow control** helps manage how fast data is sent, depending on how fast the receiver can accept it.
- TCP uses a **three-step process** before sending data:
 1. The sender asks to start a connection (**SYN**).
 2. The receiver agrees and replies (**SYN/ACK**).
 3. The sender confirms the connection (**ACK**).
- After these three steps, **data can be sent**.
- TCP uses **sequence numbers** to **keep track** of the data.
Each number goes up depending on how many bits are received, making sure the whole message is correct.

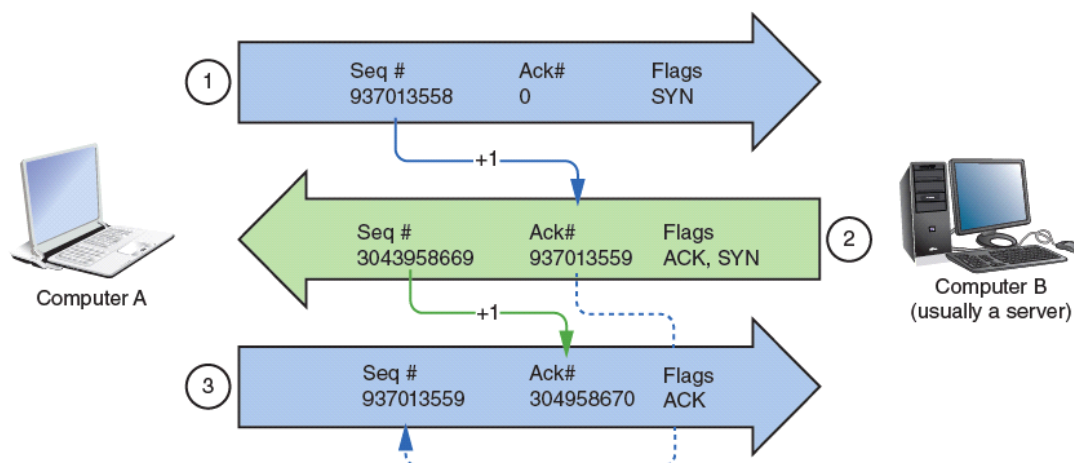


Figure 4-4 The three-way handshake process establishes a TCP session

UDP (User Datagram Protocol)

- UDP does not check for errors or the order of messages. This makes it **faster** and better than TCP in some situations.
- It is good for **live audio or video streaming**.
- UDP **works well for short messages** in one packet.
- It has only four parts: source port, destination port, length, and checksum.
- The checksum is optional in IPv4, but required in IPv6.

IP (Internet Protocol)

- IP works at the **Network layer** and decides **how and where to send data**.
- It helps TCP/IP move across different networks using routers.
- IP is **connectionless** and doesn't guarantee(**unreliable**) that data will arrive.
- It **needs TCP** to make sure the data gets to the correct place.

ICMP (Internet Control Message Protocol) !!!

- ICMP is a core protocol in the **Network layer** that helps **report success or failure of data delivery**.
- It can tell if a part of the network is busy, if data can't reach the destination, or if data was dropped because its TTL expired.
- **ICMP sends failure reports but does not fix the problems.**
- ICMPv6 does similar tasks in IPv6 as IGMP and ARP do in IPv4.

IGMP (Internet Group Management Protocol)

- IGMP works at the **Network layer** and is used to **control multicasting!**.
- **It helps with group communication** like **video calls over the Internet**.
- Routers use IGMP to find which devices are part of a multicast group and send data to them.
- Devices can use IGMP to **join or leave** multicast groups whenever they want.

ARP (Address Resolution Protocol) on IPv4 Networks !!!

- ARP helps IPv4 find the MAC address of a device on the same local network. !
- It keeps a list that matches IP addresses with MAC addresses.
- ARP works at Layer 2 but uses IP addresses from Layer 3. !
- It only works inside the **local network**. !
- ARP uses **broadcasting** to find devices. !
- The ARP table stores the IP-to-MAC address pairs.

IP Address	Hardware Address	Type
123.45.67.80	60:23:A6:F1:C4:D2	Static
123.45.67.89	20:00:3D:21:E0:11	Dynamic
123.45.67.73	A0:BB:77:C2:25:FA	Dynamic

Figure 4-12 Sample ARP table

ARP (Address Resolution Protocol) on IPv4 Networks !!!

- The ARP table has two types of entries:
 - **Dynamic:** Added when a computer sends an ARP request and gets a reply.(automatically.)
 - **Static:** Manually added by a user with the arp command.
- To check the ARP table on a Windows computer, type: `arp -a`

Routers and How They Work

- A router connects two or more networks and forwards data packets between them.
- Routers can:
 - **Connect different types of networks** like LANs and WANs.
 - Use Layer 3 and sometimes Layer 4 addresses to decide data direction.
 - **Find the best route** for data to travel from one point to another.
 - **Change the data path** if the main route is down and another one is available.

Routers and How They Work (Optional Functions)

- Routers can do more than just send data.
- They can block some broadcasts.
- They stop unwanted traffic.
- They help both local and remote connections.
- Routers use backup parts to keep the network running during failures.
- Monitor network traffic and give reports.
- Find problems and trigger alerts.

Routing Tables !!!

Routing table - a database that maintains information about where hosts are located and the most efficient way to reach them. Routers use it to decide the next router to send data to a specific host.

Routing tables contain **IP addresses** and network masks to identify networks.

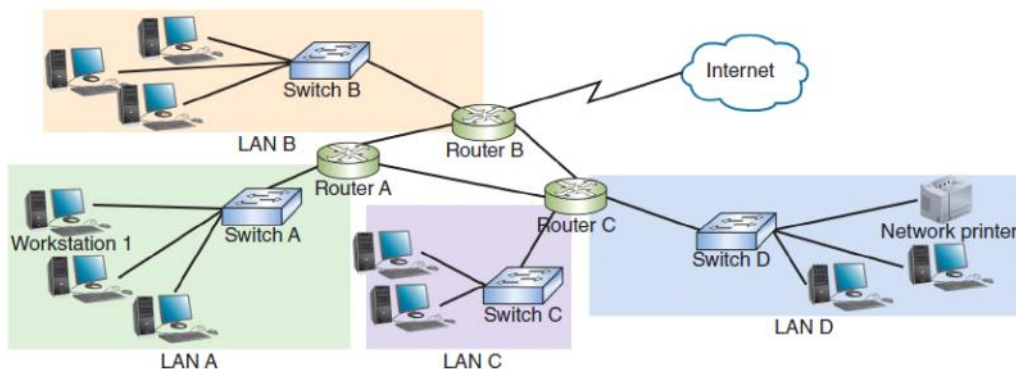


Figure 4-18 Routers rely on routing tables to locate destination hosts

Static or Dynamic Routing

- !!! • **Static routing** is set up by network admins to send data through fixed paths. (ex: small business)
- !!! • **Dynamic routing** finds the best path automatically and keep this information in a routing table. Router can **detect problems** and update routes.
- Gateway of last resort - a router that accepts all unroutable messages from other routers.

"route" Command

-The route command shows the routing table of a host.

- On Windows, use "route print" to display the table.
- On Cisco routers, use "show ip route".

Table 3-7 Fields in a routing table on a UNIX host

Destination	The destination host's identity
Gateway	The destination host's gateway
Genmask	The destination host's subnet mask
Flags	Additional information about the route, including whether it's usable (U), whether it's a gateway (G), and whether, as is the case with the loopback entry, only a single host can be reached via that route (H)
Metric	The cost of the route—that is, how efficiently it carries traffic
Ref	The number of references to the route that exist—that is, the number of routes that rely on this route
Use	The number of messages that have traversed the route
Iface	The type of interface the route uses

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Routing Metrics !!!

- Routing metrics help routers choose the best path to send data.
- They include factors like:
 - Number of hops (devices passed)
 - Speed and bandwidth of the path
 - Delay or waiting time (latency)
 - Traffic load on the path
 - Size of data packets allowed (MTU)
 - Cost assigned to the route
 - Reliability of the path
 - Network structure (topology)

Routing Protocols

- Routing protocols help routers talk to each other to find the best path for data.
- Routers check how reliable and important a protocol is by:
 - **Administrative Distance (AD):** A number showing how safe the protocol is.
 - **Convergence Time:** How fast the network finds a new best path after a change or failure.
 - **Overhead:** The amount of network resources the protocol uses.

Table 3-8 Summary of common routing protocols

Routing protocol	Type	Router location	Algorithm used
RIP (Routing Information Protocol)	IGP	Interior	Distance-vector
RIPv2 (Routing Information Protocol version 2)	IGP	Interior	Distance-vector
OSPF (Open Shortest Path First)	IGP	Interior or border	Link-state
IS-IS (Intermediate System to Intermediate System)	IGP	Interior	Link-state
BGP (Border Gateway Protocol)*	EGP	Exterior or border	Advanced distance-vector or path vector

*CompTIA classifies BGP as a "hybrid routing protocol."

Interior and Exterior Gateway Routing Protocols

- **IGP (Interior Gateway Protocols):** Used by (interior) routers inside a network system to share data and find the best paths. Two types are:
 - Distance-vector routing
 - Link-state routing
- **EGP (Exterior Gateway Protocols):** Used by border routers and exterior routers to distribute data between different network systems(outside of autonomous systems).
 - The only EGP in use today is **BGP(Border Gateway Protocol)**.

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Interior and Exterior Gateway Routing Protocols

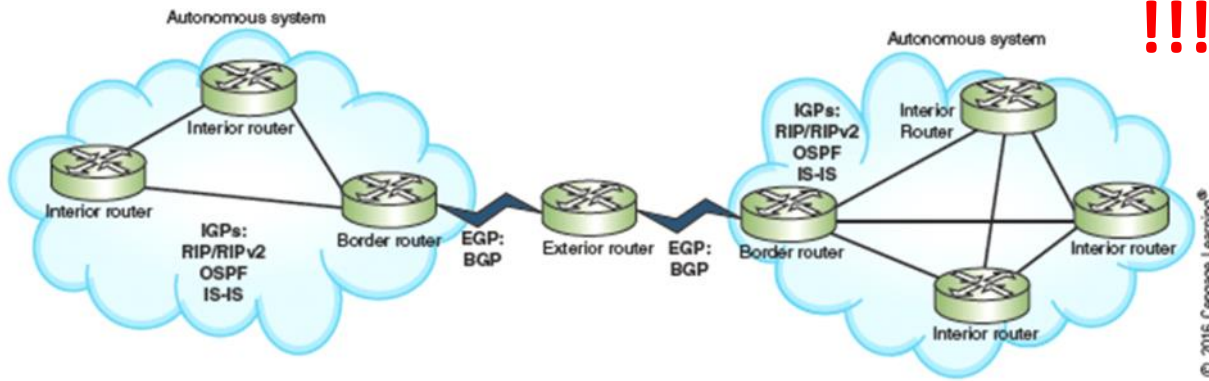


Figure 3-18 BGP is the only routing protocol that communicates across the Internet

Her sistem içinde interior (iç) ve border (sınır) router'lar IGP protokolleri (RIP, OSPF, IS-IS) ile iletişim kurarken, sistemler arası iletişim sadece BGP (Border Gateway Protocol) ile sağlanıyor. Yani, BGP internet üzerindeki farklı ağlar arasında veri alışverişini yapan tek yönlendirme protokolüdür.

OSPF (Open Shortest Path First)

- **OSPF is a type of IGP** and link-state protocol used by internal or border routers.
- It was made as a better version of RIP.
- It has no hop limit, unlike RIP.
- It uses a more complex method to find the best path.
- It needs more memory and CPU than RIP but uses less bandwidth and finds routes quickly.

IS-IS (Intermediate System to Intermediate System)

is a routing protocol used inside networks (IGP) and works similarly to OSPF.

- It uses a best-path algorithm.
- It is made for internal routers only.
- It supports both IPv4 and IPv6 easily.
- It is preferred by service providers because it can grow more easily than OSPF.

BGP (Border Gateway Protocol) is the only EGP used today and is called the “protocol of the Internet.”

- It connects **multiple autonomous systems**.
- It uses BGP messages to **share path information between routers**.
- It can follow routing rules to avoid or prefer certain paths.
- It is the most complex routing protocol.
- It uses ASN (Autonomous System Number) to recognize networks.

Summary

- **TCP** works in the Transport layer and makes sure data is delivered correctly using checksums, flow control, and sequence numbers.
- **UDP** is also in the Transport layer but does not guarantee data delivery.
- **IP** is used in the Network layer to help data go to the correct place.
- **ARP** works in the Data Link layer to find the MAC address of a device.
- **ICMP** is a Network layer protocol that shows if data is delivered successfully or not.

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!!!!!!!!!!!! (interior and border explain definitions and give examples each one of them)

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